SYED KHIZAR RAYAZ

Senior AI & Machine Learning Engineer | Agentic AI, RAG, Fintech Backends

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PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years delivering production AI for banking compliance (AML, GRC), payments and trading. Owner of 10 agentic-AI use cases in a bank-facing GRC/AML programme (3 production-ready, 2 demo-ready), built on a strict doctrine: AI drafts, deterministic code decides, humans approve, every value is source-anchored and hash-chain audited. Shipped two payment rails, an AI onboarding backend and AML regulatory reporting for a European fintech (~155 commits in 10 months), and delivered measurable outcomes including a 60% lift in onboarding conversion, a 50% cut in AML false positives and a 70% faster CI/CD pipeline. Strong in Python, FastAPI, LLM orchestration (OpenAI, Claude, Gemini, OpenRouter, Ollama), Flutter and AWS/Docker/Kubernetes. MSc Artificial Intelligence Engineering candidate (thesis track) at Istanbul Okan University.

CORE COMPETENCIES

AI/ML: Agentic AI (human-in-the-loop gates, deterministic fallbacks, hash-chained audit trails), RAG & Semantic Search, LLM Routing (OpenAI, Claude, Gemini, OpenRouter, Ollama, vLLM), Fine-Tuning & Distillation, Anomaly Detection, Computer Vision, OCR, NLP, Evaluation Harnesses, PII Masking

Frameworks & Libraries: LangChain, Hugging Face, PyTorch, TensorFlow/Keras, Scikit-learn, imbalanced-learn, OpenCV, Ultralytics/YOLO, PaddleOCR, sentence-transformers, ChromaDB, NetworkX, Streamlit

Backend & APIs: FastAPI, Django, Flask, Node.js/Express, Dart Shelf, REST & Microservices, Redis, Kafka, OAuth2/JWT, mTLS, Rate Limiting, WebSockets, Idempotent Payment APIs

Mobile & Web: Flutter/Dart (Riverpod, GetX), React Native/Expo, React, Next.js 15, Angular 19, TypeScript, PHP/CodeIgniter, Shopify Liquid, Three.js

Cloud & DevOps: AWS (EC2, S3, ECS, Textract), Docker & Compose, Kubernetes, Jenkins, CI/CD, DigitalOcean, Headless Chrome Automation

Databases: PostgreSQL, MySQL/MariaDB, MongoDB, SQL Server, ChromaDB, Elasticsearch, Prisma, SQLAlchemy

Languages: Python (primary), JavaScript/TypeScript, Dart, PHP, Java, C/C++, SQL

PROFESSIONAL EXPERIENCE

Benchmatrix

Sep 2024 - Present

AI Engineer | GRC & AML Agentic-AI Programme for Banks

- Own 10 assigned AI use cases (8 operational-risk/GRC, 2 AML) in a bank-facing agentic-AI programme; AML alert prioritization, AI case summary and transaction network analysis are production-ready, and the ScanX operational-risk and Loss Data agents are demo-ready (Nov 2026).
- Built the Loss Data Analysis Agent end-to-end (FastAPI, MySQL 8, Angular 19, OAuth2/JWT with 3 roles): 8 build stages, ~45 endpoints, 56/56 tests, SHA-256 hash-chained evidence vault; ingests 482 controls, 201 risks, 3,408 risk-control maps and 80 KRIs; deployed on Kubernetes via Jenkins for a bank demo in Oct 2026.
- Trained and deployed the AML alert-prioritization model (Random Forest, 100 trees, imbalanced-learn pipeline) and a GenAI case-analysis service with an evaluator agent and optional JWT; cut false positives by 50% (self-reported) and gave analysts a structured investigation narrative per alert.
- Benchmarked 5 local LLMs for quote-anchored extraction over real bank data (1,505 incidents, 1,081 losses, 8 dynamic form flavours): selected model reached 100% anchor survival with 0 hallucinated fields.
- Conducted senior code reviews of 3 agent codebases (KRI agent: 371 assertions; Control Breach agent: 127 checks incl. a 34-case golden harness; AML LLM-to-GBDT distillation pipeline) and documented 19 ranked findings with reproduced bugs before release.
- Rebuilt the Regulatory Requirement & Task agent with cloud/local/on-prem LLM routing (OpenAI, OpenRouter, Ollama, vLLM), JSONL audit, PII masking and a thinking trail; 32 tests run in under 1 second.
- Integrated the KRI agent (80 indicators, 327 threshold rules) into the GRC web app through a server-side proxy, and built document-forensics checks (29 modules, 51 tests) using Claude and Gemini vision models.
- Authored 33 agentic-AI use-case specifications across Incident, Loss Data and RCSA (214 pages, 66 diagrams) and a 148-note GRC knowledge vault that onboard new engineers to banking operational-risk doctrine.

- Earlier: production RAG and semantic-search pipelines for the GRC suite (ChromaDB, sentence-transformers, LangChain) and an OCR automation for a Loan Origination System on AWS Textract with GPT-4o field extraction.

OneMoneyWay Nov 2025 - Present

AI Consultant | Freelance (European fintech: wallets, cards, payments)

- Delivered ~155 authored commits across 4 repositories in 10 months (core backend 76, AI backend 60+, CRM 9, admin panel 10), adding ~30.8k lines to the core banking backend with additive, legacy-safe changes.
- Built the greenfield AI onboarding backend: per-user email cadence automation, bank-statement OCR/RAG validation (rules engine + LLM advisory with PII masking), FCM push, Basic Auth and IP rate limiting, Dockerized to AWS ECS; onboarding completion doubled.
- Built a multi-channel sales automation funnel converting registered users into onboarded customers, raising conversion by 60%, plus a RAG-powered customer-query assistant.
- Integrated two payment rails end-to-end: Treezor (wallets, cards, PCI mTLS, 3DS/SCA, WebAuthn) and Moonrise (accounts, ledger, local transfers with idempotency and locks, fee reconciliation); completed the CRM card-renewal and 3DS UI.
- Migrated communications from Bird SMS to Brevo email with activation crons, Day-7/Day-30 lifecycle rules and unsubscribe webhooks synced to contacts; produced the AML Q2-2026 regulatory export (NACE, KYB universe, batching).
- Shipped CCL statement APIs with PDF generation and as-of balance hydration, and cut onboarding query time with query optimization and new DB indexes.

Microlinks (Pvt.) Ltd Apr 2026 - Sep 2026

Software Engineer (Application Developer) | Freelance, part-time (PSX brokerage clients)

- Revamped and relaunched the FDM Securities brokerage website (CodeIgniter 4, PHP, MariaDB; 65 views, 22 controllers) with live PSX market data for 10 indices; live since 27 Jul 2026 after fixing 12 logged production issues, including a site-wide rate-limit filter and a noindex tag hiding the site from Google.
- StockBazar: ported a ~40-screen, 14.5k-line React stock-intelligence app to Flutter (28 screens, 6 platforms, offline); shipped the Express 5 marketing site and KYC onboarding module (8 document types) at stockbazar.online, and a fully local PaddleOCR CNIC/passport service with ICAO MRZ check-digit validation (1.5-3 s per scan, no cloud, no customer data retained).
- Redesigned the mobile UI of a white-label PSX trading app (6 broker builds, 832 Dart files) for Chase Securities: 17-index home, watchlist, trade sheet with PIN keypad, portfolio; v1.1.2 shipped to device and v1.3.1 delivered to testers across 30 commits.
- Rebuilt the Aryans Online Shopify store on Dawn 15.5 (57 sections, 15 templates, 42 snippets, JSON-LD structured data), live at aryans.online since 31 Jul 2026.

Alpha AI Digital Solutions Dec 2025 - Present

Software Engineer (Application Developer) | Freelance, part-time

- Designed and built PriceWise end-to-end: a React Native/Expo grocery price-comparison app that finds the cheapest stores within a 5-50 mile radius via Google Places and predicts prices with GPT-4, backed by a Node API on DigitalOcean and a Selenium price scraper; published to the Apple App Store and Google Play.
- Engineered a fully automated CI/CD pipeline for RAIS, reducing deployment effort and time by 70%.

Meteoric Solutions Jan 2024 - Aug 2024

Machine Learning Engineer | Part-time Consultant (Sep 2024 - Sep 2025)

- Spearheaded end-to-end AI initiatives from research and solution architecture through production deployment across multiple client verticals; supervised and mentored a team of AI developers.
- Built GenAI pipelines integrating OpenAI, Claude and Gemini APIs alongside local LLMs (Llama, DeepSeek), including custom fine-tuning and RAG systems.
- Deployed a RAG system for Bank Alfalah (ChromaDB, OpenAI, Elasticsearch), improving sales-team response time by 40%.
- Managed AWS infrastructure (EC2, S3) and Docker containerization; shipped mobile apps to Google Play and the Apple App Store.

Starkleytech Apr 2022 - Jun 2022

Full Stack Developer

- Architected and developed a secure, scalable cryptocurrency trading application covering end-to-end backend and frontend development.

Pakistan Petroleum Limited (PPL)

Jul 2022 - Aug 2022

Network & SAP Developer Intern

- Developed SAP reports with live Power BI integration and researched cybersecurity architectures for IT hardware procurement; recognised for outstanding performance by the IT Head.

KEY PROJECTS

- **Tafreeh (own product):** Cross-platform booking marketplace for farmhouses, beach huts and sports venues: Flutter + Dart Shelf + PostgreSQL 16, JWT auth, 3 role portals (user/owner/admin), 8 API routers, 487 seeded listings and 2,600 reviews, 10% commission and 30% advance rules; in progress.
- **Grace SRB jewellery store:** Original Next.js 15 / React 19 / Prisma e-commerce build: 53 products, 18 collections, 32 pages, cash-on-delivery checkout with server-side re-pricing, customer accounts and admin panel.
- **AML Anomaly Detection:** Ensemble model for live fraudulent-transaction detection with LLM, Ollama and Keras autoencoder backends, red-flag typology classification and promotion gates (min recall 0.90); reduced false positives by 50%.
- **Designist Patches:** Dependency-free Node static-site generator producing 69 pages from 5 data files with 8,469 verified internal links and an instant price estimator; built and paused pre-launch.
- **Polymarket paper-trading bot:** Python bot with a tested P&L ledger, arbitrage and favourite-longshot strategies and a backtester over ~90-180 resolved markets.
- **Rabta-AI:** Multilingual AI phone assistant (Faster-Whisper + Qwen 2.5 3B) handling appointment booking and SMS confirmations in Urdu and English.
- **RAG System (Bank Alfalah):** End-to-end RAG with security logging (ChromaDB + OpenAI + Elasticsearch); improved sales response time by 40%.
- **AA Joyland automation:** LLM + RAG pipeline generating CRM leads from social conversations (20% sales uplift, self-reported) and a YOLO-based maintenance-detection model scheduling repairs around peak hours.
- **JLI Edge mobile app (Jubilee):** Employee training and profiling app with CMS and reporting, deployed on iOS and Android.

EDUCATION

Istanbul Okan University, Istanbul

Sep 2026 - 2028 (expected)

MSc Artificial Intelligence Engineering (with Thesis, English) - Courses: Design of AI Systems, Image Processing, Decision Making & Robotics

University of Karachi (UBIT)

Jan 2020 - Dec 2023

Bachelor of Science in Computer Science

CERTIFICATIONS

- Oracle Cloud Infrastructure Certified Generative AI Professional
- Oracle Cloud Infrastructure Certified AI Foundations Associate

WRITING & COMMUNITY

- Technical writing on Medium and LinkedIn from real engineering work: 4 pieces published since Oct 2026 (hash-chained audit logs on MySQL, anti-anchoring design for human-in-the-loop AI agents, Flutter internals); 1,400+ LinkedIn impressions and 2,640 followers.
- Reverse-engineering case studies: static analysis of a desktop app's self-updater (2,894 words, 24 evidence files) and a document-forensics vendor blueprint.